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Can ChatGPT enhance business student creativity? Evidence from a randomised controlled trial

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ABSTRACT

This study examines the impact of generative AI (genAI) tools, specifically ChatGPT, on university student creativity across three disciplines: Marketing, Management, and Entrepreneurship. Using a randomised control trial design with 1190 undergraduate students, creativity was evaluated through fluency, originality, and elaboration. Findings reveal the use of ChatGPT had little to no effect for Entrepreneurship students yet tended to enhance creativity for Marketing and, to a lesser extent, Management students. However, its impact on originality was inconsistent, with improvements observed only in Marketing. Entrepreneurship students, with higher baseline creativity and potentially greater intrinsic motivation for the discipline, benefited less from genAI, suggesting a shift in its role from scaffolding tool to collaborative partner or productivity enhancer for students with these characteristics. The results highlight the need for educators to tailor genAI integration to disciplinary contexts and student needs. Additional research is required to explore how pedagogical factors, task and student proclivities shape genAI's effectiveness in fostering student creativity. Thoughtful integration of genAI requires well-informed educators who are prepared to develop pedagogies that harness AI's potential to complement human creativity. By fostering creative- and AI self-efficacy such pedagogical approaches can equip students to thrive in the unique demands of today's AI-driven workforce.

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Generative AI; student creativity; AI in higher education; ChatGPT; discipline specific creativity

Introduction

Today, educators stand on a burning platform. The rapid advancement of artificial intelligence (AI) technologies is reshaping the education sector in a highly uncertain and ambiguous way.

Generative AI (genAI) tools, trained on vast data sets using deep learning algorithms, can produce authoritative and persuasive human-like text, music, images, and conversations, and assist with cognitive tasks albeit not always accurately (Lin 2023). Consequently, GenAI is now positioned as a transformative force in education, offering unprecedented support for tasks critical to learning such as writing, research, and creativity (Vigeant 2024; Wang et al. 2023). However, the adoption of genAI technologies has raised significant practical and ethical concerns, including issues related to

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ethical behaviour (Yan et al. 2024), academic integrity (Birks and Clare 2023), teaching and learning practices (Kasneci et al. 2023; Tang, Chang, and Hwang 2021), and the development of creativity, an essential skill for problem-solving and innovation (Livingston 2010; Sternberg and Lubart 1998). Creativity, widely recognised as foundational for workforce readiness and economic prosperity, has recently been identified as the second most important workforce skill after analytical thinking (Khalid et al. 2020; Lubart 2001; World Economic Forum 2023), and a core competency necessary for professional success in business (Cardon et al. 2024).

The integration of genAI into the creative processes integral to learning introduces both opportunities and challenges. While AI-generated outputs challenge traditional definitions of creativity (Runco, 2023), they also open new possibilities for collaboration and exploration within educational contexts. This study focuses on the implications of genAI for the creative processes associated with university student learning, guided by the question: What are the implications of generative AI for university student creativity? Thus, we aim to investigate the impact of generative AI (specifically ChatGPT) on the divergent thinking components of creativity among undergraduate students across three academic disciplines (Marketing, Management, and Entrepreneurship.)

By examining the impact of genAI tools like ChatGPT on creativity's core dimensions, this research contributes to understanding how AI technologies might influence creativity in higher education contexts. The study provides insights into how educators may tailor the integration of genAI into pedagogical strategies to enhance creativity while addressing the unique challenges posed by genAI. In so doing, it builds on the growing body of literature on AI in education (Bozkurt 2023; Wu and Yu 2023; Yan et al. 2024) and responds to calls for empirical research on the impact of AI on creative processes (Amabile 2020). The findings contribute to the growing body of literature on AI in education by providing empirical evidence of genAI's impact on creativity across three disciplines, illustrating how genAI might support creative outputs while underscoring the importance of addressing student and task specific needs.

Literature review

Creativity in higher education

Creativity has long been recognised as a crucial skill in higher education, with its importance extending across various academic disciplines (Kleiman 2008). As a key driver of problem-solving, innovation, and personal and professional growth, creativity fosters societal and economic progress (Kleiman 2008; Livingston 2010; Moore 2007). Its importance is increasingly acknowledged as essential for workforce readiness, highlighting the need for students to develop and apply creative capabilities in their learning (Setiamurti and Kurniawati 2024).

Sternberg and Lubart (1998) define creativity as 'the ability to produce work that is both novel (i.e. original, unexpected) and appropriate (i.e. useful, adaptive concerning task constraints)' (3). In educational contexts, Amabile's (1982) componential theory of creativity provides a robust framework, identifying three key components for creative performance: domain-relevant skills (subject knowledge), creativity-relevant processes (creative thinking abilities), and task motivation (intrinsic interest and engagement). Together, these elements shape the creative potential of students.

Despite its recognised importance, the assessment of creativity remains contentious (Plucker 2023; Runco and Jaeger 2012; Said-Metwaly, Noortgate, and Kyndt 2017). Tools such as divergent thinking tests have become popular proxies for evaluating creative potential, focusing on dimensions like fluency, flexibility, originality, and elaboration (Cramond et al. 2005; Guilford 1967; Runco and Acar 2012). However, with the advent of genAI, scholars now argue for additional dimensions, such as value, authenticity, intentionality, and surprise, to differentiate human- from AI-generated outputs (Runco, 2023). Despite these advancements, traditional tools like Guilford's Alternative Uses Test (Guilford 1967) remain trusted measures for assessing creativity across disciplines (Alabbasi et al. 2022; Plucker 2023). Amabile (1982) advocates the consensual definition of

creativity, which relies on domain experts to evaluate whether a product or response is creative within its context. This subjective assessment approach continues to be a reliable method for evaluating creativity (Kaufman, Xie, and Baer 2024).

The importance of creativity is well-documented across disciplines. Jackson et al. (2006) emphasise its necessity in all fields, from the arts and humanities to science and engineering, where it drives innovation, problem-solving, and adaptability in dynamic professional environments. Within business education, creativity is a priority, with organisations like the AACSB identifying it as a critical skill for future business leaders (AACSB 2020); 71% of AACSB-accredited institutions now offer creativity and innovation courses (Zhao and Zhao 2022).

Discipline-specific research further underscores creativity's centrality. In Entrepreneurship education, creativity aligns closely with strategic thinking and opportunity recognition, fostering self-efficacy and entrepreneurial intentions (Laguía, Moriano, and Gorgievski 2019; Tantawy et al. 2021). Marketing similarly places high value on creativity, with scholars advocating for enhanced creative confidence among students to meet the profession's demands (Grant 2024; Schlee and Harich 2014; Woodall et al. 2022). Even traditionally analytical disciplines like accounting are recognising the need for creativity in exploring data and developing an innovative mindset (Mintchik, Ramamurtis, and Gramling 2021). In Management education, creativity is increasingly viewed as essential for future-focused pedagogy and student outcomes. Sunley, Harding, and Jones (2019) highlight the need to prioritise creativity in undergraduate Management courses, while Eloranta, Hakanen, and Shaw (2024) advocate for teaching radical creativity, efforts that challenge norms and foster paradigm shifts (Madjar, Greenberg, and Chen 2011).

AI and creativity

The intersection of AI and creativity has been a subject of growing interest among researchers (Anantrasirichai and Bull 2022; Elgammal et al. 2017), a key are of inquiry being whether or not these machines can truly be creative or simply mimic human creativity. Boden (1998) identified creativity as a fundamental challenge for AI, suggesting it could be overcome if AI systems could critically evaluate their own ideas thus enabling true creative transformation. Building on this, Lubart (2005) then Deterding et al. (2017) suggests that AI can serve as a collaborative partner in the creative process, emphasising the synergy between human intuition and AI's computational strengths could lead to novel combinations and perspectives. O'Toole and Horvát (2024) go further and explore how genAI tools can be integrated into human creative processes such that they facilitate new ways of 'generating, developing, and evaluating creative ideas' (6).

Other studies find evidence of the creativity of these new machines. Elgammal et al. (2017) argued they had demonstrated AI's ability to generate original artworks using the deep neural network to balance learning artistic styles whilst also deviating from them to produce novel outputs. Hubert (2024) challenges the view that creativity is a uniquely human trait, finding that even when controlling for fluency, GPT-4 demonstrated higher creative potential than did the comparative sample of humans, being both more original and elaborative in its responses to well accepted tests of divergent thinking. This finding was supported in part by Gilhooly (2024) who reviewed studies of proposed creative machines, finding that LLMs performed at near or above human levels on a test for originality and usefulness. However, Gilhooly (2024) concluded that because LLMs lack agency (i.e. they do not display intention, cannot understand their actions, do not apply judgement, and lack evaluative ability), the output of LLMs does not meet the conditions to be described creative and are better understood as purely algorithmic 'stochastic parrots' (Bender et al. 2021). Likewise, Koivisto and Grassini (2023) found chatbots did outperform humans but the best human ideas were comparable or superior to those of the chatbots, underscoring the unique and complex nature of human creativity. A similar conclusion is proposed by Anantrasirichai and Bull (2022) who review current AI technologies in use in the creative industries and conclude creative applications of AI will remain human-centric with the AI augmenting rather than replacing human creativity.

These mixed and perhaps converging outcomes not only demonstrate the promising capabilities of AI and the inherent complexity of human creativity, but also underscore the critical need for research that explores integrating AI tools into teaching practices to harness their creative potential while mitigating their limitations.

Generative AI and higher education

No matter the sector it permeates, AI has a major impact (Chen, Chen, and Lin 2020). In the education sector, large language models (LLMs) are driving the rapid evolution of educational technologies, enabling the continuous emergence of innovative tools and applications (Yan et al. 2024). The ability of LLMs to mimic human intelligence, their versatility across domains due to diverse training data, and their enhancement through reinforcement learning from human feedback (Lin 2023) position them as potentially transformative tools in higher education promising a new way of higher education teaching and learning. Recent studies are increasingly focusing on the role of AI in general and ChatGPT in particular in higher education, particularly its potential to support and enhance creativity in student learning and academic tasks (Cope, Kalantzis, and Searsmith 2021; Kasneci et al. 2023).

Given AI's capacity to enhance productivity and deliver significant gains (Dwivedi et al. 2023) there is a growing call to design a new kind of education that leverages AI as a cognitive prosthesis (Cope, Kalantzis, and Searsmith 2021). GenAI tools are being increasingly adopted to enhance efficiency, effectiveness, and the quality of learning and teaching (Chen, Chen, and Lin 2020), and meta-analyses confirm AI chatbots do have a large effect on performance, motivation, interest, self-efficacy, perceived value of learning, and anxiety with higher education students benefitting to a greater extent than students at other levels (Wu and Yu 2023). At postgraduate level the extensive use of AI tools raises the spectre that content submitted by students may not be the product of their own 'understanding or acumen/creative mind' (Segbenya et al. 2023, 9). At the same time, concerns about AI's short comings are widespread (Lin 2023), including its potential to impede learning and skills acquisition (Sáez-Velasco et al. 2024), exacerbate access inequalities (Lin 2023), produced biased outputs (Kasneci et al. 2023), and raise ethical and legal challenges (Dwivedi et al. 2023). One of the most prominent examples of genAI is ChatGPT, developed by OpenAI (Hu and Hu 2023).

Since the relatively recent advent of powerful genAI tools, literature reviews in educational settings have synthesised existing research to identify gaps, establish context, and provide a foundation for understanding and advancing knowledge on the use of generative AI in higher education. Winkler and Soellner (2018) reviewed the use of chatbots, identifying 10 key research gaps across three elements: structure (eg varying individual differences of students and educational settings), learning processes (eg applicable learning theories and the value of chatbot-based conversations) and learning outcomes (eg skills and competencies developed). Ng et al. (2022) examined 20 years of literature on AI in teaching and learning, noting a post-2021 shift from technology-focused approaches to interdisciplinary design driven by the availability of AI-powered teaching tools. Similarly, Chen, Chen, and Lin (2020) highlighted AI's continuous development and adoption in education, including its use in computing-based technologies, embedded systems, online platforms, and collaborative or independent robotics. The review by Popenici and Catalano (2023, 103) notes the dark side of edtech voicing concern over the increasing control of edtech companies as an unbalanced arrangement of power that is turning higher education into a playground of 'corporate exploitation and commercialisation of students' lives and education'. Further reviews explore AI applications within specific disciplines, such as mathematics (Hidayat et al. 2022) and pedagogical approaches such as e-learning (Tang, Chang, and Hwang 2021).

Despite these varied insights, empirical research that examines how these dynamics play out within educational settings – particularly in terms of what happens with student creativity in the

context of the AI-assisted teaching practices – is still in its early phases. This study aims to contribute towards filling that gap.

GenAI and creativity: opportunities and challenges

The collaborative potential of humans and AI in generating ideas and outputs has sparked significant interest in understanding the impact of generative AI tools like ChatGPT on creativity. Studies are exploring this intersection, highlighting both the opportunities and challenges associated with integrating AI into creative processes.

Habib et al. (2024) used the Alternative Uses Test (AUT) to evaluate ChatGPT-3's impact on student divergent thinking, measuring fluency, flexibility, elaboration, and originality. Their findings indicated that while ChatGPT enhanced creative thinking by increasing fluency, broadening flexibility, and facilitating elaboration, it also posed risks by undermining students' creative confidence by enabling cognitive fixation and reducing self-efficacy, and generating answers that were practical rather than novel or new. They advocate for the importance of carefully integrating AI tools in order to preserve the human creativity essential for learning, problem-solving, and adapting to challenges. Similarly, Sáez-Velasco et al. (2024) observe that while ChatGPT is highly regarded by educators and students in arts education for its ability to produce illustrations, its ease of use may hinder the learning of foundational art and design theories, potentially impeding the development of specialised expertise in the arts. Gao et al. (2024) also found that while ChatGPT positively influenced business students' motivation for learning and dismantled barriers to academic success, it did not enhance their self-efficacy potentially triggering imposter syndrome in the learner, highlighting the challenges in balancing AI-driven support with the need for independent learning and problem-solving.

Guzik, Byrge, and Gilde (2023) investigate ChatGPT-4 and demonstrate its exceptional performance, scoring in the top 1% for originality and fluency, alongside high scores for flexibility in a controlled study using GPT-4 submissions and undergraduate student participants. The authors propose that their findings are among the first to demonstrate AI's ability to match or surpass human capacities for original thinking, highlighting its significant potential to influence the practical applications of AI in business and everyday life. Vigeant (2024), in the context of argumentation, supports this perspective, explaining that collaborating with ChatGPT's generative capabilities facilitates human creative and critical thinking by aiding analysis, evaluation, and interpretation despite its well-known limitations. Chandrasekera, Hosseini, and Perera (2024) echoes these findings in the context of furniture design, emphasising that generative AI reduced cognitive load and acted as a co-creator in the design process, enabling the development of skill as well as enabling student abstract and lateral thinking.

Despite the growing body of research on AI and creativity, significant gaps remain in understanding how generative AI impacts the creative processes associated with learning, particularly in higher education. Creativity researchers emphasise the importance of domain-specific investigations, noting that creative processes and outputs can vary significantly across different fields of study (Plucker 2023; Runco 2014). At present, empirical studies examining how students actively engage with generative AI for creative tasks in educational settings are scarce. This gap is especially critical given the rapid adoption of AI tools like ChatGPT by students, whether deliberately integrated into curricula or informally or inappropriately used by students (Birks and Clare 2023; Lin 2023; Segbenya et al. 2023). Amabile (2020) underscores the importance of investigating the psychological and social impacts of AI on human creativity, particularly in organisational and educational contexts. Such research should not only compare AI and human creative outputs but also examine how interaction with AI tools influences students' creative processes, motivations, and learning outcomes.

While existing literature has provided valuable theoretical insights into creativity and the potential of generative AI, there remains a need for empirical studies that assess the various impacts of these tools across academic disciplines. Addressing this need is important for developing evidence-based strategies to integrate AI in education effectively while fostering the creative skills

essential for academic and professional success. Accordingly, this study employs a randomised controlled trial to investigate how the divergent thinking components of creativity differ among undergraduate business students using generative AI compared to those who do not.

Methodology

This study investigated the influence of ChatGPT Default-3.5 version (ChatGPT-3.5) on the creative output of 1190 undergraduate students studying a first year subject in three disciplines using a randomised control trial. Divergent thinking was measured under two conditions: with (experiment condition) and without (control condition) the use of ChatGPT-3.5.

Guildford's Alternative Uses Test (AUT) (Guilford 1967) is a well-established measure for divergent thinking. The test involves generating creative uses for common objects or solutions to open-ended tasks and is widely used to assess core components of creativity: fluency, flexibility, originality, and elaboration (Said-Metwaly, Noortgate, and Kyndt 2017). We chose this as a means of evaluating the impact of ChatGPT-3.5 on student creativity across the three academic disciplines because it is a domain-independent test, simple to administer, easily adaptable, and broadly acceptable as an indicator of divergent thinking across disciplines and is in use in current research in assessing the effects of ChatGPT on creative output (Haase and Hanel 2023).

The inclusion of control and experiment conditions enables direct comparisons between traditional ideation methods and ChatGPT-assisted ideation approaches. The two tasks used (listing uses for a paperclip and generating ideas to return employees to the workplace) are adaptations of the AUT that align with real-world entrepreneurial and managerial challenges. For the Entrepreneurship and Marketing students, the paper clip task is a canonical AUT variant that aligns with divergent ideation processes often used in Marketing and Entrepreneurship education (Guzik, Byrge, and Gilde 2023; Habib et al. 2024). The Management students were given a workplace-orientated task reflecting a contemporary, contextual post-Covid problem, chosen to mitigate the possibility of prior familiarity with the paperclip task identified as a potential issue for this cohort many of whom were identified as also enrolled in the Marketing subject.

Description of participants and sampling

Participants were recruited from three different first year academic disciplines, either elective or core to their degree and comprising 26 classes: Entrepreneurship ($n = 160$, 3 elective classes), Marketing ($n = 496$, 12 core classes), and Management ($n = 534$, 11 core classes). Students were randomly assigned to either the control group or the experiment (ChatGPT-3.5) group within each discipline and before their respective class. This stratified random sampling approach ensures representation across different academic disciplines while allowing for comparisons both within and between disciplines despite any disciplinary differences. During data cleansing, duplications (ie students attending two subjects) were removed.

Measures and procedures

Participants were divided into two groups – control and experiment – within each of the three academic disciplines. In the Control Group, students generated ideas independently using their usual resources, such as discipline materials, personal knowledge, or internet searches. They were explicitly instructed not to use any generative AI tools, including ChatGPT, during the task. In the Experiment Group, students were provided access to ChatGPT-3.5 and were specifically instructed to interact with the tool to assist in generating their ideas for the given task. Before the data collection commenced, all students were given guidance on using ChatGPT-3.5 for ideation while maintaining focus on the creative output. This design enabled a direct comparison between traditional, unaided ideation processes and those enhanced by ChatGPT-3.5 assistance.

The tasks were conducted under standardised conditions to ensure fairness and consistency. To ensure comparability of results the same protocol was followed by each of the five researchers who attended the 26 classes to invite participation in this ethics approved research project and collect the signed hardcopy consent forms. Students were given 15 minutes to complete their respective tasks.

All participants received identical verbal and written instructions prior to task commencement including watching a 2-minute instructional video on how to prompt ChatGPT-3.5, but otherwise developed their own prompts according to their understanding. The task was administered by a researcher who was not the classroom teacher. Responses were submitted to the Learning Management System (LMS) via a non-graded item on the LMS that repeated the same instructions delivered by the researcher in-class and provided a submission site. Procedural consistency ensured that any observed differences in creative output could be attributed to the use of ChatGPT-3.5 rather than external factors.

The subjective assessment technique was used to obtain reliable judgements of creativity (Amabile 1982). The task meets the requirements for this technique because it is not heavily reliant on expertise or skills of the students to complete; the raters all had practical and theoretic experience of the three domains; and the tasks were consistent with real work evaluations of creative responses enabling the appropriate raters to determine creativity. To further strengthen the reliability and validity of the assessment, the three raters used a rubric with definitions for each indicator, adapted from widely accepted creativity literature that captures diverse aspects of divergent thinking as a measure of potential for creative thinking (Runco and Acar 2012; Said-Metwaly, Noortgate, and Kyndt 2017). Prior to rating, the raters met online to review and discuss the marking rubric used to assess flexibility, originality and elaboration. This step ensured a shared understanding of the criteria, which were rated on a 5-point scale from Exceptional to Insufficient. Each rater independently scored the same submission and discussed their reasoning to align interpretations. This process was repeated six times, moderating 12 submissions (six for the paperclip task and six for the return-to-office task), resulting in a consensus on how to apply the rubric categories. The key creativity measures were chosen based on their widespread use in creativity research and their ability to capture different aspects of divergent thinking, a recognised indicator of the potential for creative thinking (Runco and Acar 2012):

(1) Fluency (the ability to generate a large quantity of ideas within a limited amount of time), a count of the total number of ideas generated in the task by each respondent.
(2) Flexibility (the ability to shift perspectives or approaches in response to changing situations or constraints), measured as the variety of different categories of ideas produced.
(3) Originality (the ability to generate novel or unique ideas or approaches), measured using the rubric criterion of uniqueness of ideas compared to what each rater saw in the overall sample.
(4) Elaboration: (the ability to provide detail and complexity to ideas), measured as the degree of detail, specificity, and richness provided for ideas (Said-Metwaly, Noortgate, and Kyndt 2017).

Student submissions to the LMS were downloaded into a spreadsheet, with incomplete data and submissions from students who did not consent to participate in the research removed from the sample. The refined datasets were then distributed to each rater, who iteratively reviewed all responses. Raters recorded the fluency count and used the rubric to evaluate flexibility, originality and elaboration. Each rater assessed the submissions from all participating students.

Pearson correlations were used to assess consistency in scoring across the three raters. We found strong inter-rater agreement for the Fluency ($rs = .83–.95$, $ps < .001$) and Flexibility ($rs > .75–.88$, $ps < .001$), and comparatively lower agreement for Elaboration ($rs = .60–.66$, $ps < .001$) and Originality ($rs = .32–.49$, $ps < .001$). In a MANOVA, when independent variables (or factors) are highly correlated, they tend to explain nearly identical variance across the dependent variables. Since MANOVA evaluates group differences on a combination of measures, the inclusion of highly correlated predictors

introduces redundancy – eliminating one factor generally does not affect the multivariate test statistics (e.g. Wilks' Lambda, Pillai's Trace).

Given that fluency and flexibility produce essentially identical outcomes in our analysis, reporting fluency is theoretically justified. Fluency serves as the primary index of creative ideation and is widely recognised as the foundational process in divergent thinking (Guilford 1967; Runco and Acar 2012). Established theoretical frameworks, such as Guilford's model of creativity, position fluency as the core cognitive mechanism from which other dimensions, including flexibility, are derived (Guilford 1967). Emphasising fluency not only provides a more direct measure of creative potential but also simplifies the operationalisation of our constructs by reducing subjectivity in categorising responses. This focus enhances the methodological clarity and parsimony of our study.

Results

To assess the impact of the intervention, two Multivariate Analysis of Variance (MANOVA) were run. The first MANOVA was a 2 (Discipline: Entrepreneurship vs Marketing) by 2 (Group: Intervention vs Control), with the components of creativity entered as the dependent variables. A separate MANOVA was run for the Management discipline due to the use of a different creativity task. As such, the second MANOVA included Discipline (Entrepreneurship vs Marketing) as the only IV.

Although an overall comparison between experimental and control groups could provide insights into the general effect of generative AI on creativity, our primary focus was to explore discipline-specific differences. Preliminary analyses did not reveal significant overarching trends beyond those observed within each discipline, justifying our emphasis on disciplinary comparisons.

Data pre-processing

A series of Pearson correlations indicated that the assumption of multicollinearity was violated, with a very strong correlation identified between flexibility and fluency. As a result, the flexibility variable was dropped from the analysis. Data screening revealed that residuals for the three remaining outcome variables were significantly non-normal for both MANOVA analyses. After removing outliers, visual inspection of the Q-Q plots suggested no further issues with normality of residuals. Moreover, Mahalanobis distances revealed two multivariate outliers ($\chi^2(4) = 18.46$, $p < .001$), which were subsequently removed. No additional univariate outliers were identified. Curve estimation indicated that the assumption of linearity was met, with linear relationships identified across the three variables. However, Box's M test was significant for the first MANOVA, indicating that the variance-covariance matrices were not equal across groups. Although MANOVA is reasonably robust to violations of the assumption of homogeneity of variance-covariance matrices, results need to be interpreted with caution. For this analysis, the Pillai's trace results were reported. The assumption was met for the second MANOVA.

Does ChatGPT impact creativity?

Entrepreneurship and marketing students

Multivariate results indicated a significant main effect of intervention (Pillai's trace = 0.06. $F(3, 196) = 4.63$, $p = .004$), partial $\eta^2 = .06$), a significant main effect of discipline (Pillai's trace 0.19. $F(3, 196) = 15.31$, $p < .001$, partial $\eta^2 = .19$), and a significant interaction between intervention and discipline (Pillai's trace = .09, $F(3, 196) = 6.75$, $p < .001$, partial $\eta^2 = .09$). Follow-up univariate tests revealed no significant differences overall in fluency ($F(1,198) = 0.14$, $p = .70$) or elaboration ($F(1, 198) = 0.001$, $p = .974$) between the intervention and control groups. The intervention group scored significantly higher in originality than the control group ($F(1,198) = 25.62$, $p < .001$).

Table 1. Creativity measures as a function of intervention group and disciplines.

	Control	Intervention	Significance
Fluency			
Entrepreneurship	3.22 (0.23)	3.11 (0.23)	.752
Marketing	2.52 (0.16)	3.65 (0.20)	<.001
Originality			
Entrepreneurship	1.49 (0.11)	1.45 (0.11)	.817
Marketing	0.79 (0.08)	1.14 (0.09)	.005
Elaboration			
Entrepreneurship	1.18 (0.10)	1.12 (0.10)	.643
Marketing	0.80 (0.07)	1.50 (0.09)	<.001

Table 2. Creativity measures across intervention group for management.

	Control	Intervention	Significance
Fluency	1.78 (0.08)	2.29 (0.09)	<.001
Originality	0.66 (0.04)	0.77 (0.04)	.064
Elaboration	1.06 (0.10)	1.99 (0.11)	<.001

Results also revealed significant interactions between intervention and discipline for fluency ($F(1, 198) = 8.89$, $p = .003$, partial $\eta^2 = .04$) and elaboration ($F(1, 198) = 17.23$, $p < .001$, partial $\eta^2 = .08$), but no significant interaction for originality ($F(1, 198) = 3.80$, $p = .053$, partial $\eta^2 = .019$).

Pairwise comparisons showed no significant differences between the intervention and control groups in the Entrepreneurship discipline across any of the outcomes. In contrast, in the Marketing discipline, the intervention group scored significantly higher than the control group across all outcomes. See Table 1 for descriptive and inferential statistics.

Management students

Multivariate results indicated a significant main effect of intervention (Wilks Lambda = .07, $F(3, 118) = 19.92$, $p < .001$), partial $\eta^2 = .33$.

Follow-up univariate tests revealed significant differences in fluency ($F(1,120) = 18.69$, $p < .001$, partial $\eta^2 = .14$) and elaboration ($F(1, 120) = 41.34$, $p < .001$, partial $\eta^2 = .26$) with the intervention group scoring significantly higher than the control group. In contrast, there no significant difference in originality across the groups ($F(1,198) = 3.50$, $p = .06$, partial $\eta^2 = .03$). Descriptive statistics are displayed in Table 2.

Discussion

This study examined the implications of genAI (specifically ChatGPT-3.5) for university student creativity. We conclude that genAI enhances university student creativity by scaffolding fluency and elaboration, particularly in abstract, ideation-focused tasks, but its impact on originality maybe context-dependent. We suggest that in contexts where creativity skills are highly effective and intrinsic task motivation is strong, or where creativity is deeply intertwined with personal insight and contextual judgment, the value proposition of generative AI shifts from being a foundational support tool to serving as a collaborative partner or productivity enhancer.

Overall, our findings reveal that while participants who used ChatGPT-3.5 generally scored higher in creativity compared to those in the control condition, this effect varied depending on the students' discipline and the specific component of creativity being measured. Notably, ChatGPT-3.5 use had little to no impact on creativity among Entrepreneurship students but tended to enhance creativity among Marketing students and, to a lesser extent, Management students. Furthermore, the effect of ChatGPT-3.5 was dependent on the particular measure of creativity employed, indicating that AI tools may not uniformly or consistently affect the different components of creativity. Importantly, students in the intervention groups across all disciplines demonstrated high levels of

engagement with ChatGPT-3.5, reflecting a broad willingness to integrate AI tools into their academic work.

Our findings align with existing research that finds that genAI can facilitate divergent thinking by enabling rapid idea generation and developing detailed, expansive outputs (Chandrasekera, Hosseini, and Perera 2024; Guzik, Byrge, and Gilde 2023). In considering why there were differences between disciplines, we speculate by drawing upon Amabile's componential model of creativity (Amabile 1983) to suggest the interplay of domain-relevant skills, creativity-relevant processes, and intrinsic motivation may have shaped these findings.

Our students were drawn from two core (Marketing and Management), and one elective (Entrepreneurship) subject. We did not collect data that would allow us to know if the students in the core subjects were in these subjects to fulfil their degree obligation (i.e., one subject out of eight required 1st year business subjects) or as part of their discipline related major sequence (i.e., one out of eight discipline related subjects). We know many of these students will be fulfilling degree requirements to study one only subject in each of the Marketing and Management disciplines, and suggest this in turn could point to lower intrinsic motivation and less-developed domain or creativity relevant skills (Amabile 1983) in that subject. ChatGPT-3.5 likely compensated effectively for such gaps by scaffolding creativity, particularly in Marketing, where the alignment between the assigned task and genAI's strengths maximised its impact.

In contrast, all the Entrepreneurship students will have self-selected into this subject. Research suggests entrepreneurs rely on intuitive decision-making, opportunity recognition, and the ability to connect non-obvious ideas (Baron 2006; Gillin and Sinclair 2020; Sarasvathy 2008). Consistent with research (Laguía, Moriano, and Gorgievski 2019) and Amabile's model (1983) we suggest students who elect to study entrepreneurship may exhibit higher intrinsic motivation and pre-existing creative skills. Notably, the baseline results for Entrepreneurship students in the control group were higher across all indicators (fluency, originality, and elaboration) compared to both Management and Marketing students. Thus we suggest it is when intrinsic motivation is high, and creativity skills are already well-developed, ChatGPT's utility changes. However, as intrinsic motivation and individual baseline creativity were not explicitly measured in this study, this interpretation remains speculative and a limitation. Future research should incorporate validated instruments to directly assess these traits (see for example Karwowski, Lebuda, and Wiśniewska 2018).

The study also highlighted the limits of genAI in fostering originality, a critical dimension of creativity. Although the inter-rater reliability for the originality measure was low, our findings make theoretical sense. While Marketing students showed significant improvements in originality, this was not observed in the Management students whose task was consistent with the cognitive demands of management tasks which are diverse, often involve synthesising ideas from various domains and navigating complex human dynamics (Urban et al. 2024a). ChatGPT's strengths in generating expansive ideas were evident, but its limitations in originality support calls for human evaluative input to refine and innovate upon AI-generated outputs (Chandrasekera, Hosseini, and Perera 2024; Deterding et al., 2017; O'Toole and Horvát 2024; Vigeant 2024). What is not explored in this research is whether the results tended towards conventional ideas that did not sufficiently push the boundaries of originality (Guzik, Byrge, and Gilde 2023; Habib et al. 2024), or whether our evaluators perceiving the AI-assisted ideas (as identified based on sheer quantity of responses) viewed them as less unique.

These disciplinary differences suggest the importance of aligning genAI applications with the specific creative demands of each academic field. We do recognise the use of the classic paperclip task for Marketing and Entrepreneurship versus a return-to-workplace task for Management may have contributed to variations in scores. Given that the paperclip task is known to be associated with divergent thinking (Habib et al. 2024), differences in outcomes might be partly attributable to task characteristics rather than solely to disciplinary factors.[1] However, it is worth noting that the management students' use of AI yielded similar effects to Marketing students despite the use of different tests, whereas, Entrepreneurship and Marketing had largely different findings despite using the same test. This suggests that findings did not appear to vary as a function of the type of creativity test used.

Nonetheless, AI has been found to enhance creativity (O'Toole and Horvát 2024) particularly in ideation-focused disciplines (Grech, Mehnen, and Wodehouse 2023), therefore its value proposition for intuition-driven or highly contextual domains may be different. This finding underscores the criticality of incorporating theoretical grounding in the process of researching creativity in order to analyse and substantiate how student creativity is fostered (Setiamurti and Kurniawati 2024).

Finally, the integration of AI tools like ChatGPT into education presents a paradox for educators; it both empowers and undermines student learning. Thus, educators' understanding of genAI tools like ChatGPT is critical. Familiarity with these tools enables educators to explore pedagogical applications, mitigate risks, and design strategies tailored to diverse student needs. For highly motivated and creative students, genAI can be a collaborator refining ideas and streamlining tasks, while for less motivated students or those with foundational skill gaps, it offers scaffolding to engage with tasks they might otherwise avoid. Leveraging genAI strategically can foster student confidence, competence, and engagement levels (Wu and Yu 2023) which in turn may lead to the development of intrinsic interest where it may be lacking.

By fostering both creative and AI self-efficacy in students, educators can equip students to critically engage with genAI as a collaborative tool that serves as a catalyst for growth rather than a substitute for their abilities or crutch (Chee, Ahn, and Lee 2024; Karwowski 2015). Aligning genAI with both disciplinary and student needs will ensure genAI enhances creativity without undermining human capacities. Furthermore, in designing AI-centric pedagogies educators will need to be cognzient of the self-efficacy challenges inherent in using AI: fostering over-reliance on the technology, or reducing confidence in their own abilities, or driving overconfidence in own abilities (Gao et al. 2024; Habib et al. 2024; Urban et al. 2024).

Revising our pedagogical approaches to effectively integrate genAI into our teaching and learning will not only prepare learners to meet the technological demands of a rapidly evolving workforce (World Economic Forum 2023) but also safeguard the authenticity and intentionality that define human creativity (Runco, 2023). In doing so, higher education can strike a balance between leveraging AI's potential and nurturing the uniquely human capacities that are essential for innovation and problem-solving.

Conclusion

This research highlights the potential and challenges of genAI, specifically ChatGPT-3.5, for university student creativity. While the use of ChatGPT-3.5 significantly enhances fluency and elaboration in some disciplines, its impact on originality was inconsistent and seemingly varied by task and student proclivities. The findings underscore the need for educators to tailor pedagogical approaches to balance AI-enhanced outputs with activities that foster self-efficacy and intrinsic creative motivation. Existing creativity models, such as Amabile's (1982) componential model, should be expanded to incorporate AI's role as a collaborator in creative processes. Further research is needed to examine how pedagogical practices can be tailored to influence genAI's capacity to support student creativity. The interplay between task type and whether a subject is core or elective, the intrinsic motivation of students by discipline, and the impact of AI on originality, offer avenues for future research.

The study is limited by a lack of generalisability. Additionally, individual differences in prior experience with AI and prompt engineering, a lack of standardisation with respect to prompt quality, the nature of the tasks themselves, and the subjective assessment of creativity potentially affected the outcomes. Addressing these limitations and gaps in future research will enhance our understanding of the impact of genAI on student creativity.

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